

**Question 1:**

**(12 marks)**

- (a) Express 0.04678 in scientific notation correct to 3 significant figures. [2]
- 

- (b) Rationalise the denominator of  $\frac{3+\sqrt{2}}{2-\sqrt{2}}$ . [3]

Simplify your answer as far as possible.

- (c) Solve for  $\theta$  if  $2\cos\theta=1$  and  $0^\circ \leq \theta \leq 360^\circ$ . [3]

- (d) Solve for  $k$  if  $3(k-1)-2(k-2)>1$ . [2]

- (e) Express the recurring decimal  $0.\dot{2}\dot{2}$  as a fraction in its lowest terms. [2]

Question 2: *Start a new sheet of paper*

(12 marks)

- (a) Find the derivatives of each of the following with respect to  $x$ :

(i)  $x^3 - 5x^2 + 7x + e^3$

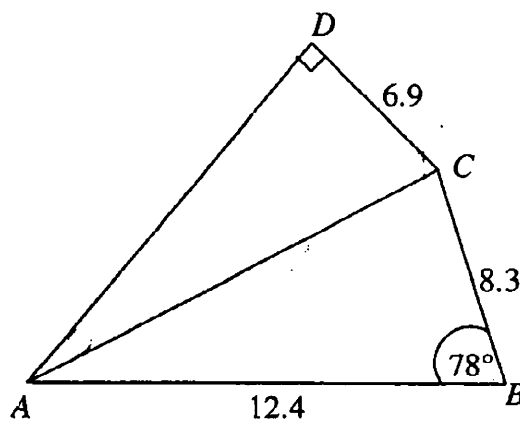
(ii)  $\ln(x^2 + 5)$

(iii)  $\tan(3x)$

(iv)  $\frac{\sin x}{x^3}$

[6]

- (b) The quadrilateral  $ABCD$  is shown in the diagram – *all measurements are in centimetres*.



(Diagram not to scale)

- (i) Find, using the Cosine Rule, or otherwise, the length of  $AC$ , correct to one decimal place.
- (ii) Find the size of  $\angle CAD$ , giving your answer to the nearest degree.
- (iii) Calculate, correct to one decimal place the area of quadrilateral  $ABCD$ .

[6]

**Question 3:**      *Start a new sheet of paper*

**(12 marks)**

A circle  $C$ , with centre  $(-4, 2)$  is tangent to the straight line  $l$  with the equation  $3x + 4y - 16 = 0$ . The perpendicular distance from the centre of this circle to the straight line  $l$  is equal to the radius of the circle.

- 
- (a) Find the  $x$  and  $y$  intercepts,  $A$  and  $B$ , respectively of the line  $l$ . [2]
- (b) Show that the radius of the circle,  $C$ , is 4 units. [3]
- (c) Write down the equation of the circle,  $C$ , in the form  
$$(x - h)^2 + (y - k)^2 = r^2$$
 [1]
- (d) On a suitable number plane, sketch the circle,  $C$  and the straight line,  $l$ . [2]
- (e) Verify that the coordinates of the point of intersection,  $E$ , of the circle,  $C$ , and the line,  $l$  is  $(-1.6, 5.2)$ . [1]
- (f) Find the equation of the line,  $k$ , that passes through  $(4, -2)$ , and is also perpendicular to the line  $l$ . [2]
- (g) Verify that the circle,  $C$ , intersects the  $y$  axis at the point  $D$ , with coordinates  $(0, 2)$ . [1]

Question 4:      *Start a new sheet of paper*

(12 marks)

- (a)    (i)    Find the number of terms there are in the arithmetic series

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$$258 + 251 + 244 + \dots - 288 - 295 ?$$

- (ii)    Calculate the sum of the arithmetic series in (a)(i) above.      [4]

- (b)    (i)    Find the discriminant of the quadratic equation

$$x^2 - (2 - k)x + 1 = 0.$$

- (ii)    Hence, or otherwise, determine the values of  $k$  for which  
the graph of

$$y = x^2 - (2 - k)x + 1$$

cuts the  $x$  – axis in two distinct points.      [4]

- (c)    A parabola's equation is given by  $y = x^2 - 2x - 3$ .

- (i)    Find the focal length.  
(ii)   Determine the coordinates of the vertex  $V$ .  
(iii)   Find the coordinates of the focus  $S$ .      [4]

Question 5: *Start a new sheet of paper*

(12 marks)

- (a) The common ratio  $r$  of a geometric progression satisfies the quadratic equation

$$2r^2 - 3r - 2 = 0.$$

- (i) Solve for  $r$

If the sum to infinity of the same progression is 6,

- (ii) Explain why, in this case,  $r$  can only take on one value.  
Hence, state the common ratio  $r$ .

- (iii) Show that the first term  $a$  of this progression, is 9. [5]

- (b) A population of eels in a certain river has become too large to be supported by its environment and it decreases from time  $t = 0$  to a stable level.

The size of the population in thousands of eels is given by  $g(t)$ , where  $t$  is measured in weeks.

The rate of change of the population is given by  $g'(t) = -5e^{-2t}$  for  $t \geq 0$ . It is known that when  $t = 0$ ,  $g = 10$ .

- (i) Find, by integrating  $g'(t) = -5e^{-2t}$ , the size of the population of eels at any time  $t$ .
- (ii) Sketch the graph of  $y = g(t)$  on a number plane.
- (iii) State the long-term stability level of the eel population in the river. [7]

Question 6: *Start a new sheet of paper*

(12 marks)

(a) Consider the function  $f(x) = x^3 - x^2 - x + 1$ .

- 
- (i) By using the *grouping in pairs method for factorising*, or *otherwise*, factorise fully the expression  $x^3 - x^2 - x + 1$ .
- (ii) Evaluate  $f(1)$  **and**  $f(-1)$ .
- (iii) Hence, or otherwise, state the  $x$  intercepts of  $f(x)$ .
- (iv) Determine the coordinates **and** the nature of the stationary points on  $y = f(x)$ .
- (v) Find the coordinates of the point(s) of inflexion of  $y = f(x)$ .
- (vi) Hence, sketch the graph of  $y = f(x)$  on a number plane, showing all essential features. [8]

(b) Forty five percent of a population are of blood group  $O$ , 40% are of blood group  $A$  and the remainder are neither group  $O$  nor group  $A$ .

*Three people are chosen at random from the population.*

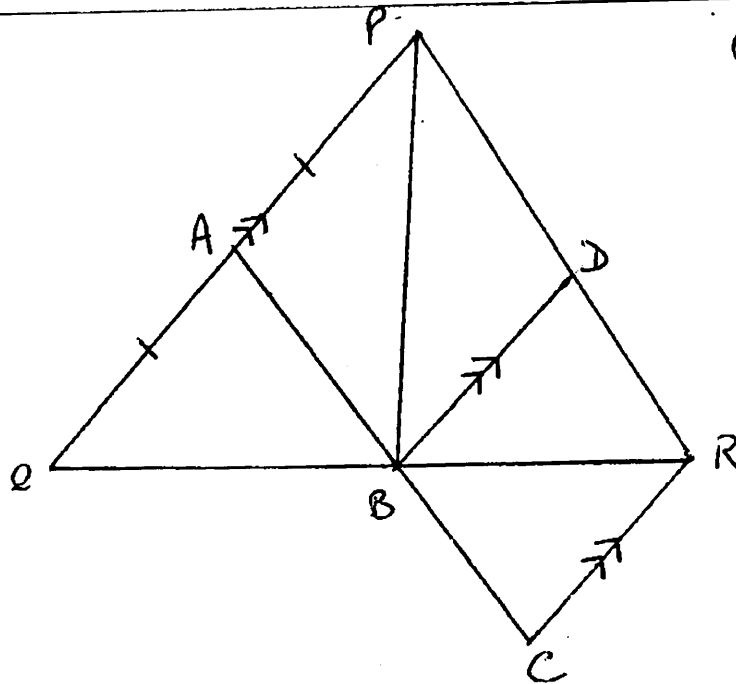
By drawing a tree diagram, or otherwise, find the probability that

- (i) all three people are of blood group  $A$ ;
- (ii) two of the people are of blood group  $A$  and the other is group  $O$ ;
- (iii) there is one person each of group  $O$ , group  $A$  and neither group  $O$  nor group  $A$ . [4]

Question 8: *Start a new sheet of paper*

(12 marks)

In triangle  $PQR$ ,  $B$  is a point on  $QR$  such that  $PB$  is the angle bisector of angle  $RPQ$ .  $A$  is the midpoint of  $QP$ . The straight line joining  $AB$  is produced to a point  $C$  such that  $CR \parallel QP$ .



(Diagram is not to scale)

*Copy this diagram onto your answer sheet.*

- (a) Prove, with reasons, that  $\triangle BAQ$  is similar to  $\triangle BCR$ . [3]

- (b) Hence, state why  $\frac{RC}{QA} = \frac{RB}{QB}$ . [1]

- (c)  $BD$  is a line drawn parallel to  $QP$  meeting  $PR$  at  $D$ .

Show that  $\frac{RD}{PD} = \frac{RB}{QB}$ . [2]

- (d) By showing  $\triangle BRD$  is similar to  $\triangle QRP$ , or otherwise, show that

$$\frac{RD}{PD} = \frac{PR}{PQ}. \quad [3]$$

- (e) Hence, or otherwise, show that  $RC = \frac{1}{2}PR$ . [3]

Question 7: *Start a new sheet of paper*

(12 marks)

- (a) (i) On a suitable number plane sketch the graph of  $y = |x - 1|$ .

- 
- (ii) Solve the inequality  $|x - 1| < 1$ .

- (iii) Hence, or otherwise, evaluate  $\int_0^2 |x - 1| dx$ . [4]

- (b) (i) Find  $\int \frac{1}{2x - 5} dx$ .

- (ii) Find  $\int_0^{\frac{\pi}{3}} \cos(6x) dx$ , leaving your answer in simplified form. [4]

- (c) Let  $A$  be the area enclosed by the curve  $y = \sqrt{x} + \frac{1}{\sqrt{x}}$ , the  $x$  axis and the lines  $x = 4$  and  $x = 9$ .

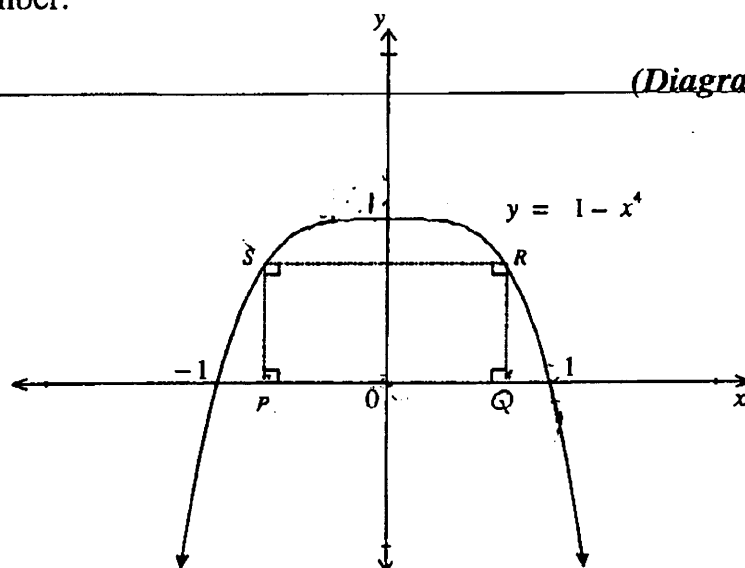
Find the volume of the solid formed by rotating this area through  $360^\circ$  about the  $x$  axis.

*Leave your answer in simplified form and in terms of  $\pi$ .* [4]

**Question 9:**      *Start a new sheet of paper*

**(12 marks)**

The diagram shows part of the graph of the function  $y = 1 - x^4$ , where  $x$  is any real number.



- (a) Calculate  $I$ , the area enclosed between the graph of  $y = 1 - x^4$  and the  $x$  axis. [3]

- (b) The rectangle  $PQRS$  has coordinates

$$P(-u, 0), Q(u, 0), R(u, y(u)), \text{ and } S(-u, y(-u))$$

where  $0 < u < 1$ .

Let the area of the rectangle  $PQRS$  be denoted by  $A(u)$ .

- (i) State why  $A(u) = 2u(1 - u^4)$  square units.
- (ii) Verify that  $A(0.5) = \frac{15}{16}$ .
- (iii) Let  $u_1$  be the value of  $u$  for which  $A(u)$  is a maximum. Find the value of  $u_1$  correct to five significant figures.
- (iv) Calculate the maximum value of  $A(u)$ , giving your answer correct to 5 significant figures.
- (v) Show that if  $A(u) = uI$ , then  $u = u_1$ . [9]

**Question 10:**      *Start a new sheet of paper*

(12 marks)

Dusty Lifesaver is retiring from the work force. Dusty has a lump sum (from a superannuation fund) of \$ $P$  to invest. Dusty is advised that it will take \$ $m$  to cover living expenses in any given year after retirement with the remainder invested at  $r\%$  per annum and compounded annually. The living expense of \$ $m$  is taken out of the remainder at the beginning of each year.

Let  $R = 1 + \frac{r}{100}$  and assuming  $P > R(P - m)$  is satisfied.

(a) Show that

(i) after 1 year Dusty has \$  $R(P - m)$  remaining.

(ii) after 2 years Dusty has \$  $PR^2 - mR(1 + R)$  remaining.

(iii) after 3 years Dusty has \$  $PR^3 - mR(1 + R + R^2)$  remaining. [3]

(b) Show that  $1 + R + R^2 = \frac{R^3 - 1}{R - 1}$ . [1]

(c) By first observing the pattern established in (a), show that after  $n$  years the remaining superannuation in dollars is given by

$$PR^n - \frac{mR(R^n - 1)}{R - 1}. \quad [2]$$

(d) By first equating the expression in (c) to zero, show that Dusty's superannuation is *exhausted* when

$$R^n = \frac{mR}{P - R(P - m)}. \quad [3]$$

(a) Hence, or otherwise, find the number of years Dusty can survive if  $P = \$400\,000$ ,  $m = \$40\,000$  and  $r = 7\%$  per annum. [3]

**END OF EXAMINATION**

Question 1

a)  $4.68 \times 10^{-2}$

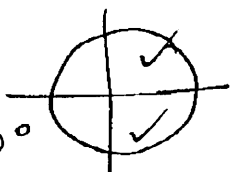
(2)

b)  $\frac{3+\sqrt{2}}{2-\sqrt{2}} = \frac{(3+\sqrt{2})(2+\sqrt{2})}{(2-\sqrt{2})(2+\sqrt{2})}$   
 $= \frac{6+5\sqrt{2}+2}{4-2}$   
 $= \frac{8+5\sqrt{2}}{2}$

(3)

c)  $2\cos\theta = 1 \quad 0^\circ \leq \theta \leq 360^\circ$   
 $\Rightarrow \cos\theta = \frac{1}{2}$

$\Rightarrow \theta = 60^\circ \text{ or } 300^\circ$



(3)

d)  $\Rightarrow 3(k-1) - 2(k-2) > 1$

$\Rightarrow 3k-3-2k+4 > 1$

$\Rightarrow k+1 > 1$

$\Rightarrow k > 0$

(2)

e)  $\frac{0.\dot{2}\dot{2}}{99} = \frac{22}{99}$   
 $= \frac{2}{9}$

(2)

Question 2

a) i)  $3x^2 - 10x + 7$

ii)  $\frac{2x}{x^2+5}$

iii)  $3\sec^2 3x$

iv)  $\frac{u'v - uv'}{v^2} = \frac{\cos x \cdot x^3 - \sin x \cdot 3x^2}{x^6}$

$= \frac{x \cos x - 3 \sin x}{x^4}$

$x \neq 0$

(6)

b) i)  $AC^2 = 12.4^2 + 8.3^2 - 2 \times 12.4 \times 8.3 \times \cos 78^\circ$   
 $= 179.853 \quad (3 \text{ dp}) \quad \text{COS RULE}$

$\Rightarrow AC = \underline{13.4 \text{ correct to 1 dp}}$

ii)  $\angle CAD = \sin^{-1} \frac{6.9}{AC}$

$= \underline{31^\circ} \text{ to nearest degree}$

iii)  $\text{Area} = \text{Area } \triangle ADC + \triangle ACB$

$= \frac{1}{2} AC \times CD \sin \angle DCA$   
 $+ \frac{1}{2} AB \times BC \times \sin 78^\circ$

$= \frac{1}{2} \times 13.41 \times 6.9 \times \sin 59^\circ$

$+ \frac{1}{2} \times 12.4 \times 8.3 \times \sin 78^\circ$

$= 36.66 + 50.34$

$= \underline{90.0 \text{ cm}^2} \quad (1 \text{ dp})$

(6)

### Question 3

a)  $l: 3x + 4y - 16 = 0$

$x$  intercept  $(\frac{16}{3}, 0)$

$y$  intercept  $(0, 4)$

②

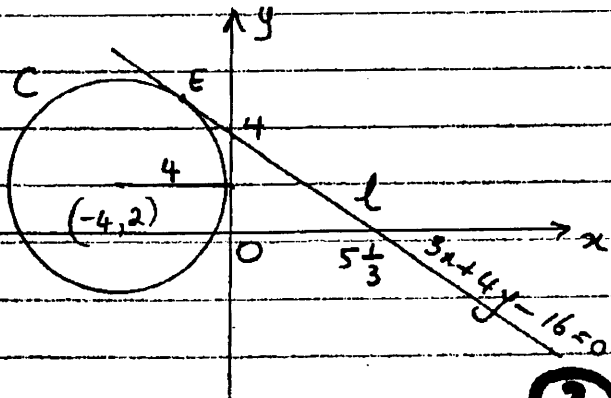
b)  $r = d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$   
 $= \frac{|3(-4) + 4(2) - 16|}{\sqrt{3^2 + 4^2}}$   
 $= \frac{20}{5} = 4$

③

c)  $(x+4)^2 + (y-2)^2 = 4^2$

①

d)



②

e)  $E(-1.6, 5.2)$  sub in

$l: 3(-1.6) + 4(5.2) - 16$

$= -4.8 + 20.8 - 16$

$= 0 \checkmark$  lies on line  $l$ .

$C: (-1.6+4)^2 + (5.2-2)^2$

$= 2.4^2 + 3.2^2$

$= 16 = r^2 \checkmark$  lies on circle

$\therefore E$  is on both  $\therefore E$  is the intersection.

①

f)  $m_l = -\frac{3}{4}$

$M_k = \frac{4}{3} \quad (4, -2)$

$y + 2 = \frac{4}{3}(x - 4)$

$3y + 6 = 4x - 16$

$4x - 3y - 22 = 0$

②

g)  $(0, 2)$  lies on the x-axis.

sub  $(0, 2)$  in  $C$ :

$(0+4)^2 + (2-2)^2 = 16 + 0$

$= 16 \checkmark$  lies on  $C$

$\therefore (0, 2)$  is intersection  $D$

①

### Question 4

a) i)  $a = 258, d = -7$

$T_n = -295 = a + (n-1)d$

$= 258 + (n-1)(-7)$

$= 258 - 7n + 7$

$\Rightarrow 7n = 560$

$\Rightarrow n = 80$

ii)  $S_n = \frac{n}{2}(a + l)$

$= \frac{80}{2}(258 - 295)$

$= -1480$

④

b) i)  $\Delta = b^2 - 4ac$

$= (-2-k)^2 - 4(1)(1)$

$= 4 - 4k + k^2 - 4$

$= k(-4 + k)$

ii)  $\Delta > 0$  for 2 distinct pts

i.e.  $k(k-4) > 0$

i.e.  $k < 0$  and  $k > 4$

④

c) i)  $y = x^2 - 2x - 3$   
 $= (x-1)^2 - 1 - 3$   
 $\Rightarrow y+4 = (x-1)^2$   
 $|a| = \frac{1}{4}$

ii)  $V = (1, -4)$

iii)  $S = (1, -3\frac{3}{4})$

(4)

### Question 5

i)  $2r^2 - 3r - 2 = 0$

$\Rightarrow (2r+1)(r-2) = 0$

$r = -\frac{1}{2}$  or  $r = 2$

If the sequence has a sum to infinity  
 $|r| < 1$  i.e.  $r = -\frac{1}{2}$

i)  $S_{\infty} = \frac{a}{1-r} = 6$

$a = 6 \times (1 - -\frac{1}{2})$

$a = 9$

(5)

b) i)  $g'(t) = -5e^{-2t}$   
 $\Rightarrow g(t) = \frac{5}{2}e^{-2t} + C$

$g(0) = 10$

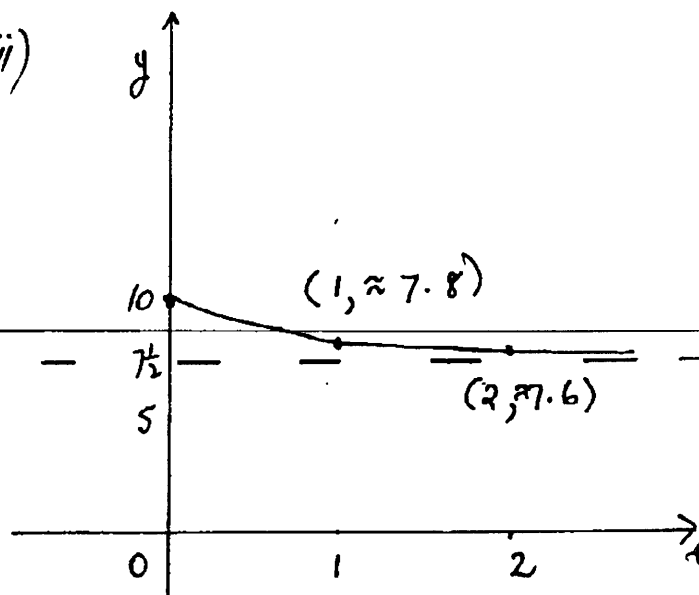
$\Rightarrow 10 = \frac{5}{2}e^{-2 \cdot 0} + C$

$\Rightarrow 10 = 2\frac{1}{2} + C$

$\Rightarrow C = 7\frac{1}{2}$

$\Rightarrow g = \frac{5}{2}e^{-2t} + 7\frac{1}{2}$

ii)



iii) 7500 eels

(7)

### Question 6

a) i)  $x^2(x-1) - 1(x-1)$   
 $= (x^2-1)(x-1)$

$= (x-1)(x+1)(x-1)$

ii)  $f(1) = 0$ ,  $f(-1) = 0$

iii)  $(1, 0)$ ,  $(-1, 0)$

iv) For stationary pt  
 $f'(x) = 0$

1.  $3x^2 - 2x - 1 = 0$

2.  $(3x+1)(x-1) = 0$

$\Rightarrow x = -\frac{1}{3}$  or  $x = 1$

$f''(x) = 6x - 2$

$x = -\frac{1}{3}$   $f''(x) < 0$  max

$x = 1$   $f''(x) > 0$  min.

ctd

MAX at  $(-\frac{1}{3}, \frac{32}{27})$

MIN at  $(1, 0)$

v) Non stationary inflection of

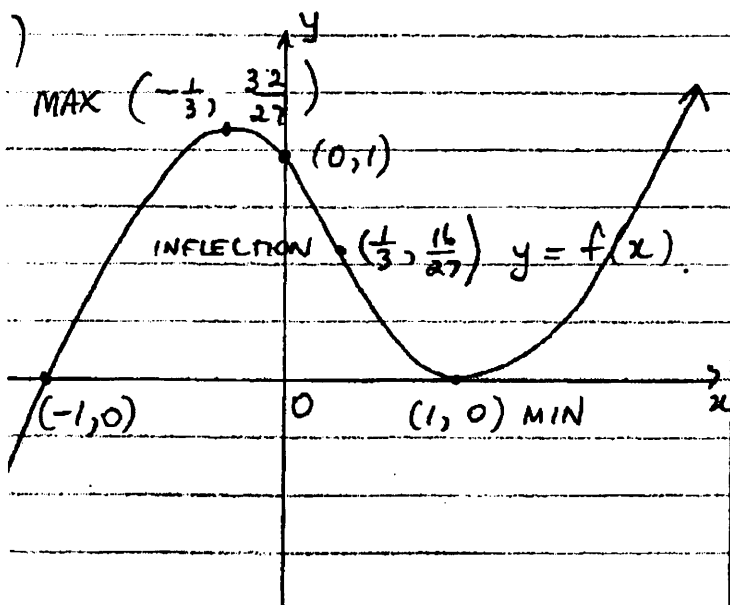
$$f''(x) = 0 \text{ (at } f'(x) \neq 0)$$

$$\text{i.e. } 6x - 2 = 0$$

$$\text{i.e. } x = \frac{1}{3}$$

$$\Rightarrow y = \frac{16}{27}$$

Coords  $(\frac{1}{3}, \frac{16}{27})$



$$i) P(0) = 45\% \quad P(A) = 40\%$$

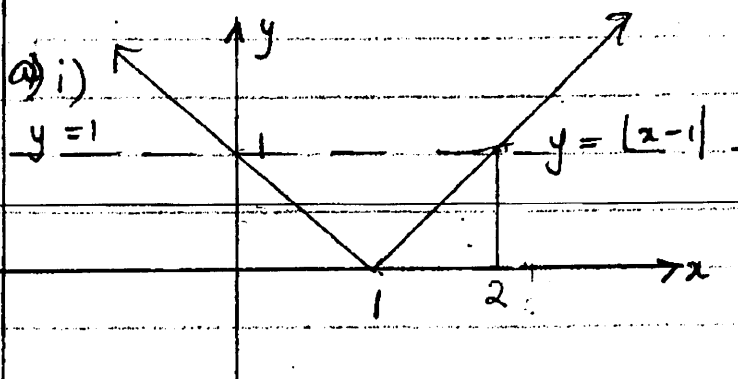
$$i) P = (0.4)^3 = 0.064$$

$$ii) P = 3 \times (0.4)^2 \times 0.45 = 0.216$$

$$iii) P = 6 \times 0.4 \times 0.45 \times 0.15 = 0.162$$



Question 7



$$ii) 0 < x < 2$$

$$iii) \int_0^2 |x-1| dx = \text{area under } y = |x-1| \text{ for } 0 < x < 2$$

$$= 2 \times \frac{1}{2} \times 1 \times 1$$

$$= 1$$



b)

$$i) \int \frac{1}{2x-5} dx = \frac{1}{2} \ln|2x-5| + C$$

$$ii) \int_0^{\pi/3} \cos 6x dx = \left[ \frac{\sin 6x}{6} \right]_0^{\pi/3}$$

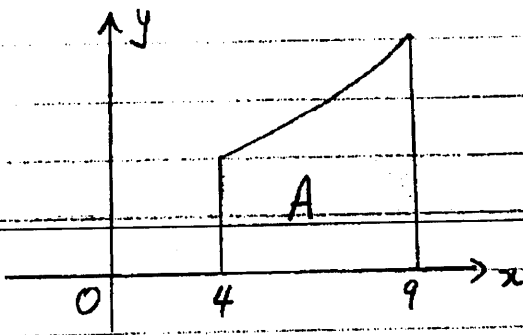
$$= \frac{\sin 2\pi}{6} - \frac{\sin 0}{6}$$

$$= 0$$



7 ctd

c)



$$V = \pi \int_a^b y^2 dx$$

$$= \pi \int_4^9 \left( \sqrt{x} + \frac{1}{\sqrt{x}} \right)^2 dx$$

$$= \pi \int_4^9 \left( x + 2 + \frac{1}{x} \right) dx$$

$$= \pi \left( \frac{x^2}{2} + 2x + \ln x \right) \Big|_4^9$$

$$= \pi \left( \frac{81}{2} + 18 + \ln 9 - \left( \frac{16}{2} + 8 + \ln 4 \right) \right)$$

$$= \pi \left( \frac{85}{2} + \ln \frac{9}{4} \right)$$

$$\text{OR } \pi \left( \frac{85}{2} + 2 \ln \frac{3}{2} \right)$$

Question 8

a) In  $\triangle BAP$ ,  $\triangle BCR$

$\angle PAB = \angle RCB$  (alt  $\angle$ 's  $PA \parallel CR$ )

③  $\angle PBA = \angle RBC$  (vert opp  $\angle$ 's)

$\Rightarrow \triangle BAP \parallel \triangle BCR$  (equiangular)

b) corresponding sides of

① similar  $\triangle$ 's are in proportion

c)  $BD \parallel CP \Rightarrow$

②  $\frac{RD}{PD} = \frac{RB}{PB}$  (intercept theorem)

d) In  $\triangle BCD$ ,  $\triangle CDP$

$\angle R$  is common

$\angle DBR = \angle PPR$  (corr  $\angle$ 's  $CP \parallel BD$ )

$\Rightarrow \triangle BCD \parallel \triangle CDP$

$\Rightarrow \frac{RD}{BD} = \frac{RP}{CP}$  (corresponding sides)

But  $\angle DBP = \angle APB$  (alt  $\angle$ 's  $AP \parallel B$ )  
and  $\angle DPB = \angle APB$  (given)

$\Rightarrow \angle DBP = \angle DPB$

$\Rightarrow \triangle PBD$  is isosceles

$\Rightarrow BD = PD$

$\therefore \frac{RD}{PD} = \frac{RP}{CP}$  from \* ③

e) Proven  $\frac{RC}{PA} = \frac{RB}{PB}$ ,  $\frac{RD}{PD} = \frac{RB}{PB}$

$\Rightarrow \frac{RC}{PA} = \frac{RD}{PD}$

but  $\frac{RD}{PD} = \frac{RP}{CP}$

$\Rightarrow \frac{RC}{PA} = \frac{RP}{CP}$

$\Rightarrow \frac{RC}{PR} = \frac{PA}{PP} = \frac{1}{2}$  ( $A = \text{midpt } PQ$ )

$\Rightarrow RC = \frac{1}{2} PR$  as required. ③

### Question 9

$$a) \int_{-1}^1 (1-x^4) dx = \left[ x - \frac{x^5}{5} \right]_{-1}^1 = 2\left(1 - \frac{1}{5}\right) = \frac{8}{5} \quad \textcircled{3}$$

$$\Rightarrow 1-u^4 = \frac{4}{5} \quad (0 < u < 1)$$

$$\Rightarrow u^4 = \frac{1}{5}$$

$$\Rightarrow u = \sqrt[4]{\frac{1}{5}}$$

$$b) i) \text{Area} = l \times b$$

$$A(u) = 2u \times f(u)$$

$$= 2u \times (1-u^4)$$

$$ii) A(0.5) = 2(0.5) \times (1-0.5^4) = 1 - \frac{1}{16} = \frac{15}{16}$$

$$iii) \frac{dA}{du} = \frac{d}{du} (2u - 2u^5)$$

$$= 2 - 10u^4$$

$$= 0 \text{ for stationary pt}$$

$$\Rightarrow u^4 = \frac{1}{5}$$

$$\Rightarrow u = \sqrt[4]{\frac{1}{5}}$$

$$= 0.66874 \text{ to 5 sig figs}$$

Test nature of SP

$$\frac{d^2A}{du^2} = -40u$$

$$< 0 \text{ for } u = \sqrt[4]{\frac{1}{5}}$$

$$H. \text{ max S.P. at } u_1 = 0.66874 \text{ (sf)}$$

$$iv) A(u_1) = 2\left(\sqrt[4]{\frac{1}{5}}\right) \left(1 - \frac{1}{5}\right)$$

$$= \frac{8}{5} \left(\sqrt[4]{\frac{1}{5}}\right)$$

$$= 1.06998$$

$$= 1.0700 \text{ to 5 sig fig.}$$

$$v) A(u) = uI$$

$$\Rightarrow 2u(1-u^4)^{\frac{1}{4}} = uI$$

$$= u \times \frac{8}{5}$$

$$= u, \text{ (see iv above).} \quad \textcircled{9}$$

### Question 10

a) i) After 1 year:

$$A_1 = (P-m)(1+r\%)$$

$$= (P-m)R$$

$$= \text{A (P-m) remaining}$$

ii) After 2 years

$$A_2 = ((P-m)R - m)R$$

$$= PR^2 - mR^2 + mR$$

$$= PR^2 - mR(1+R)$$

iii)  $A_3$  after 3 yrs

$$A_3 = \{[PR^2 - mR(1+R)] - m\}R$$

$$= PR^3 - mR^2(1+R) - mR$$

$$= PR^3 - mR(R^2+1) - mR$$

$$= PR^3 - mR(1+R+R^2) \quad \textcircled{3}$$

$$b) (1+R+R^2)(R-1)$$

$$= R - 1 + R^2 - R + R^3 - R^2$$

$$= R^3 - 1$$

$$\Rightarrow 1+R+R^2 = \frac{R^3-1}{R-1} \quad \textcircled{1}$$

c) Following the pattern in a)

$$A_n = PR^n - mR(1+R+R^2+\dots+R^{n-1})$$

$$= PR^n - mR \frac{(R^n-1)}{R-1} \quad \textcircled{2}$$

d) If  $A_n = 0$

$$PR^n = \frac{mR(R^n - 1)}{R - 1}$$

making  $R^n$  the subject:

$$\Rightarrow R^n(PR - P) = mR R^n - mR$$

$$\Rightarrow R^n(PR - P - mR) = -mR$$

$$\Rightarrow R^n = \frac{mR}{P - RP + Rm}$$
$$= \frac{mR}{P - R(P - m)} \quad (3)$$

e)  $\$P = \$400\,000$   $\$m = \$40\,000$

$$r = 7\%$$

$$\Rightarrow 1.07^n = \frac{40000(1.07)}{400000 - 1.07(360000)}$$

$$\Rightarrow 1.07^n = \frac{42800}{14800}$$
$$= 2.89189...$$

$$\Rightarrow n = \frac{-\log(2.89189...)}{\log 1.07}$$
$$= 15.7 \text{ yrs (3 sig fig).}$$

(3)